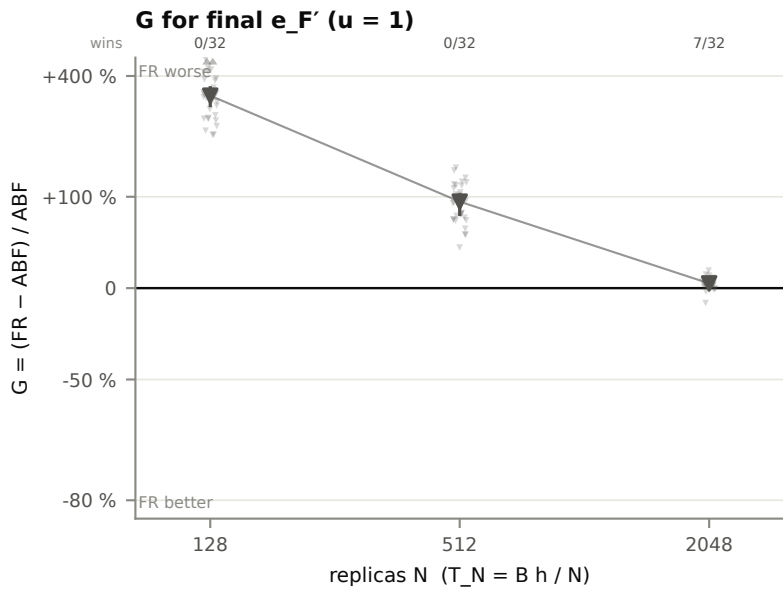
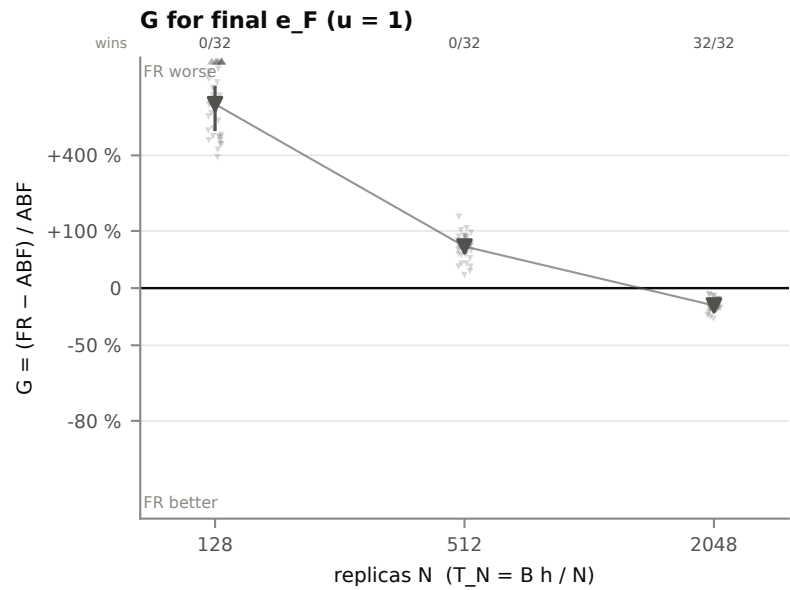
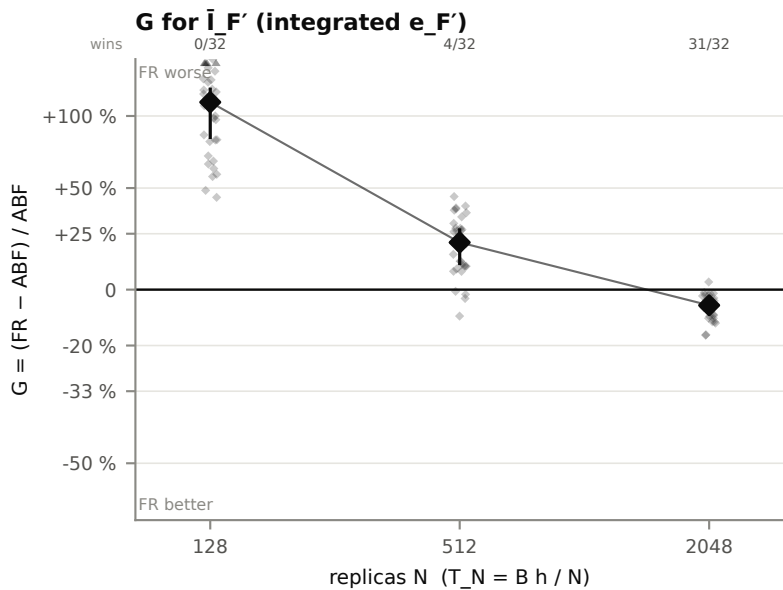
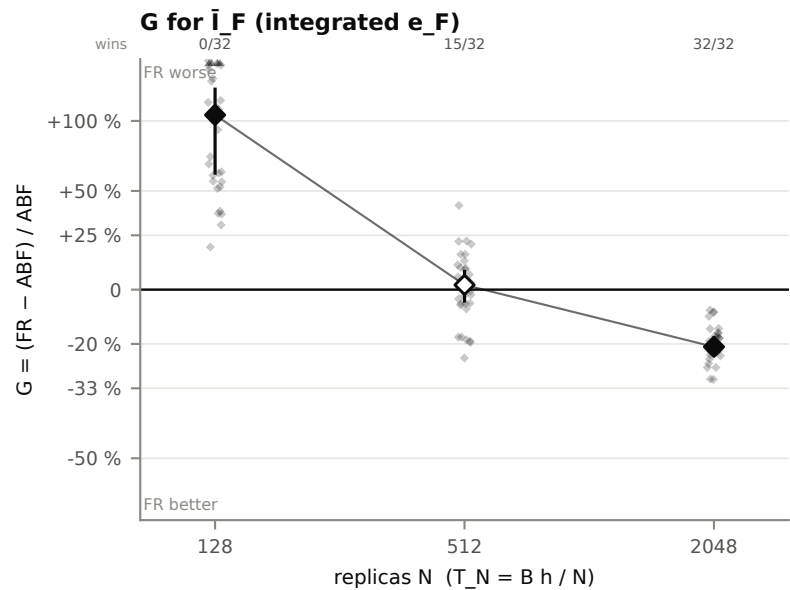


# gateway shift: FR relative improvement G(N) at equal budget

Equal total budget per arm per seed:  $B = N \times n\_steps = 3.2768e+09$  walker-steps at every  $N$  ( $h = 2.5e-05$ ,  $T\_N = B h / N$ ); 32 seeds per ( $N$ , method).  
Paired per seed (same  $N$  and seed: shared initial conditions and Langevin noise).



gateway shift · S3\_fr\_gain\_vs\_N · status: all planned ( $N$ , seed, method) complete · FR is not defined at  $N = 1$  (ABF only, by design).  
Marker: median of the per-seed  $G$ ; bar: seed-bootstrap 95 % CI of the median (10000 resamples); filled = CI excludes 0; faint: per-seed  $G$  (carets: off scale). Log-ratio axis (FR/ABF) labelled in % change. 'wins' row: seeds with  $FR < ABF$  / paired seeds. Red 'FR inactive' / 'no death  $k/n$ ': FR arms that realised no death (equal to ABF), where  $G = 0$  is FR inactivity, not equivalence.  
Source results/mechanism/matched\_free\_energy/shift/analysis/summary.json (generated 2026-10-10T14:19:18Z, eqb\_metrics/2, 10000 bootstrap resamples); plot\_synthesis/2.